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LMI characterization of structural and robust stability: the discrete-time case [☆]

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Abstract

This paper extends to the discrete-time case some robust stability conditions, recently obtained for continuous-time systems. Those conditions are expressed in terms of Linear Matrix Inequalities (LMI), being thus simply and efficiently computable. As in the continuous-time case, parameter-dependent Lyapunov functions can be constructed and, consequently, the new approach can yield much sharper and less conservative results than the simultaneous stability approach. In particular, well-known stability problems, namely, D-stability and robust stability in the presence of diagonally structured uncertainty can be more efficiently addressed. Numerical examples are included to illustrate the advantages of the new stability conditions. © 1999 Elsevier Science Inc. All rights reserved.

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1. Introduction

In a recent paper [1], new structural robust stability conditions for continuous-time systems were derived. Those conditions, which consist in verifying the asymptotic stability of all elements of a given convex set of matrices, are significantly less conservative, and provide much sharper stability results than other known methods with similar computational complexity. The aim of this paper is to derive the discrete-time counterpart of these results. We will show that, closely following the approach of [1] and introducing appropriate modifications in the analysis of the discrete-time case, a complete extension of the continuous-time case can be achieved.

Before proceeding to the main body of this paper, some concepts have to be defined. A real matrix $A \in \mathbb{R}^{n \times n}$ is said to be Schur stable if all its eigenvalues are located in the interior of the open unitary disk in the complex plane. It is well known that matrix A is Schur if, and only if, there exist two symmetric and positive definite matrices P and Q satisfying the linear constraint $A'PA - P + Q = 0$, called discrete-time Lyapunov equation. In the affirmative case, the equilibrium point $x = 0$ of the time invariant linear difference equation $x(k+1) = Ax(k)$ is asymptotically stable. Furthermore, if the above Lyapunov equation admits a *positive diagonal* solution P , it is possible to ensure global asymptotic stability to the equilibrium point $x = 0$ of the nonlinear difference equation $x(k+1) = Af(x(k))$, where $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ represents any function belonging to some well defined set [2,3], a fact which was first established, in the continuous-time case, by Persidskii [4]. Thus, the existence of a diagonal P means robust stability with respect to the set of allowed nonlinearities which includes linear functions of the form $f(x) = Bx$ where B belongs to the convex polyhedron

$$\mathcal{B}_D := \{B = \text{diag}(b_1, \dots, b_n) : |b_i| \leq 1, i = 1, \dots, n\}. \quad (1)$$

Therefore, the existence of a positive diagonal solution to the Lyapunov equation implies that any matrix of the form AB , $B \in \mathcal{B}_D$ is Schur stable. In general, the converse of this statement is not true. This fact is connected to a problem hitherto unsolved for both continuous-time [5] or discrete-time [3] systems, which consists in completely characterizing the class of D-stable matrices. In the discrete-time setting this class is composed by all matrices A such that the product AB is stable for all $B \in \mathcal{B}_D$.

No difficulty is faced in testing whether some given matrix A is Schur or not. Given any positive definite matrix Q , this is accomplished by numerically solving the discrete-time Lyapunov equation, a well-studied problem with efficient algorithms available. In the case P is required to be positive diagonal, cutting-plane [6] and LMI [7] algorithms have also been developed. For D-stability the situation is quite different and, to our knowledge, no simple numerical procedure is available. One possible approach to the D-stability

problem is to use simultaneous Lyapunov functions, i.e., try to find a single and constant matrix P which proves the stability of the whole set of matrices. This, however often leads to highly conservative results. The method to be described in this paper allows matrix P to be parameter dependent, thus reducing the conservativeness of the stability condition.

1.1. Notation

The notation used throughout the paper is mostly standard. Capital letters denote matrices and small letters denote vectors. The closed convex polytope defined by a finite number of extreme matrices, say $A_1, \dots, A_N$, is defined as the convex hull of that element and is denoted simply as $\text{conv}_N(A)$. Given a convex set $\mathcal{C}$ and $A_i \in \mathcal{C}$ for all $i = 1, \dots, N$ then $\text{conv}_N(A) \subset \mathcal{C}$ for all $N > 0$. For a finite N , $\text{conv}_N(A) = \mathcal{C}$ is called a convex *polyhedron* [8]. Moreover, any convex set can be arbitrarily approximated by a convex polyhedron, for sufficiently large N . The subscript D in a matrix, like P_D , means that it is constrained to be diagonal. For matrices and vectors, we use $(\cdot)'$ to denote transposition and $P > 0$ to denote that P is symmetric and that its eigenvalues are strictly positive. Finally, for square matrices, $\text{trace}(\cdot)$ denotes the trace of $(\cdot)$.

2. Problem statement and definitions

In this paper, we consider the following set of square matrices

$$\mathcal{A} := \{AB : B \in \mathcal{B} \subset \mathbb{R}^{n \times n}\}, \quad (2)$$

where $A \in \mathbb{R}^{n \times n}$ is a given matrix and $\mathcal{B}$ is a fixed and known convex (not necessarily polyhedral) set. As an extension of the continuous-time case results of [1], our main purpose is to provide conditions under which all matrices belonging to $\mathcal{A}$ are Schur. In the affirmative case, we simply say that $\mathcal{A}$ is Schur. Hence this paper deals with the following robust stability problem “Given $(A, \mathcal{B})$ determine whether $\mathcal{A}$ is Schur or not”.

Several known stability problems may be recast in this formulation. Indeed, for the trivial cases $\mathcal{B} = \{I\}$, $\mathcal{A} = \{A\}$ the problem above reduces to the classical stability problem of a single matrix A . If $A = I$ and $\mathcal{B} = \text{conv}_N(B)$, then we are testing the stability of all matrices in $\mathcal{A} = \mathcal{B}$, the same problem considered, for instance, in [9] using the simultaneous stability framework. Finally the D-stability problem is formulated by defining $\mathcal{B} = \mathcal{B}_D$ given by (1).

3. Stability conditions

Throughout this section we assume that $\mathcal{B} = \text{conv}_N(B)$. In this form, given the N extreme matrices B_i , $i = 1, \dots, N$ every matrix in the set $\mathcal{B}$ is recovered by

$$B(\xi) := \sum_{i=1}^N \xi_i B_i, \quad \xi_i \geq 0, \quad \sum_{i=1}^N \xi_i = 1. \quad (3)$$

From the previous discussion our goal is to determine a *parameter dependent* quadratic Lyapunov function

$$v(x, \xi) := x' P(\xi) x \quad (4)$$

such that

$$v(x(k+1), \xi) - v(x(k), \xi) := B(\xi)' A' P(\xi) A B(\xi) - P(\xi) < 0 \quad (5)$$

for all $\xi_i \geq 0$, $\sum_{i=1}^N \xi_i = 1$. The existence of such a function proves that the origin $x = 0$ of the set of difference equations $x(k+1) = ABx(k)$ with $B \in \mathcal{B}$ is globally stable. Although the determination of a matrix $P(\xi)$ as a function of ξ is, in general, a very difficult task, for continuous-time systems the Extended Strictly Positive Real (ESPR) property provided a way to overcome this difficulty. The main idea was to build an auxiliary system with dynamic matrix BA and to prove that testing this system for the ESPR property is indeed equivalent to testing the original matrix AB for stability. A key property used in the proof of this equivalence (see [1]) is that if AB is Hurwitz, so A and B are both full rank matrices. In the discrete-time case, although this is no longer true (even the zero matrix is Schur!) we can still state an equivalent result with the help of the following theorem.

Theorem 1. *Matrix AB is Schur stable if, and only if, there exist matrices $P > 0$, $G \in \mathbb{R}^{n \times n}$ and $H \in \mathbb{R}^{n \times n}$ satisfying the LMI*

$$\begin{bmatrix} GB + B'G' - P & B'H' - G \\ HB - G' & A'PA - H - H' \end{bmatrix} < 0. \quad (6)$$

Proof. To prove sufficiency we assume that there exists $P > 0$ so that (6) is feasible. Then, multiplying (6) by the matrix $[I \ B']$ on the left and by its transpose on the right we get

$$B'A'PAB - P < 0,$$

which implies that matrix AB is Schur stable. To prove the necessity we assume that AB is Schur stable and solve the discrete-time Lyapunov equation

$$B'A'PAB - P = -Q < 0$$

to get $P > 0$. Then we set

$$G = 0, \quad H = A'PA + hI \quad (7)$$

and choose $h > 0$ sufficiently small such that

$$h^{-2}(A'PA + 2hI) > BQ^{-1}B'$$

holds. Noticing that Q is strictly positive definite we apply Schur complement formula (see [1]) to the above inequality and replace the values of G , H and Q to obtain

$$\begin{bmatrix} B'A'PAB - P & B'A'PA - G - B'H \\ A'PAB - G' - H'B & A'PA - H - H' \end{bmatrix} < 0. \quad (8)$$

The proof is concluded by defining the nonsingular matrix

$$M := \begin{bmatrix} I & 0 \\ B & I \end{bmatrix}$$

and noticing that (8) is equal to

$$M'(*)M < 0, \quad (9)$$

where $(*)$ replaces the left-hand term of inequality (6). Hence, from the non-singularity of M we can draw that given AB Schur stable there always exists a sufficiently small $h > 0$ such that LMI (6) holds. This concludes the proof. $\square$

The most important point in the stability condition given above is that the LMI (6) does not involve the product of variables B and P as in the standard Lyapunov inequality. In fact, matrices P and B appear linearly in LMI (6), which is the key property for the stability results involving sets of matrices to be derived.

Also notice that, as in the continuous-time case [1], it is possible to interpret condition (6) in the light of the discrete-time analog [10,11] of the continuous-time ESPR property. In fact, since (6) is similar to (8), it can be shown that testing the stability of matrix AB is completely equivalent to testing the existence of matrices G and H so that the transfer function

$$T(z) := (G' + H'B)(zI - AB)^{-1}A + H \quad (10)$$

is ESPR. Although that equivalence holds for matrices G and H which are not constrained in any manner, the necessity part of the above proof was carried with the help of the very particular matrices G and H given in (7), which suggests that the given stability condition is overparametrized as far as stability of a single matrix AB is concerned. This fact will be explored in order to reduce the conservatism of the forthcoming stability conditions.

Theorem 2. For $\mathcal{B} = \text{conv}_N(B)$ the following properties are equivalent:

- (i) The set of matrices $\mathcal{A}$ is simultaneously stable, that is, there exists a constant matrix $P > 0$ such that $B'A'PAB - P < 0$, $\forall B \in \mathcal{B}$.
- (ii) There exist matrices $P > 0$, G and H satisfying the LMI

$$\begin{bmatrix} GB_i + B_i'G' - P & B_i'H' - G \\ HB_i - G' & A'PA - H - H' \end{bmatrix} < 0, \quad i = 1, \dots, N. \quad (11)$$

Proof. The proof of this theorem follow the same steps as in [1]. We prove first that (i) implies (ii). Looking at the proof of Theorem 1 and assuming that (i) holds then

$$-Q_i := B_i'A'PAB_i - P < 0$$

for some $P > 0$. Now defining matrices G and H as in (7) and choosing h sufficiently small such that

$$h^{-2}(A'PA + 2hI) > B_iQ_i^{-1}B_i' \quad (12)$$

holds for all $i = 1, \dots, N$ we can ensure that inequality (8) holds for all i if B is replaced by B_i . Hence, following the same steps as in the proof of Theorem 1 for each value of i we conclude that (11) holds with (7) and (12). To prove the converse, from (3) we multiply each inequality in (11) by ξ_i and add all terms, which yields

$$\begin{bmatrix} GB(\xi) + B(\xi)'G' - P & B(\xi)'H' - G \\ HB(\xi) - G' & A'PA - H - H' \end{bmatrix} < 0.$$

Now it suffices to multiply the above inequality to the left by $[I \ B(\xi)']$ and to the right by its transpose to get (5) which is satisfied by the constant P for all $\xi_i \geq 0$, $\sum_{i=1}^N \xi_i = 1$ and concludes this proof. $\square$

From the proof of Theorem 2 it is also apparent that matrices G and H may be constrained without introducing any additional conservatism. In other words, the stability condition (ii) of Theorem 2 is still overparametrized as far as simultaneous stability is concerned. Moreover, as in Theorem 1, if (11) is feasible then it is possible to verify that the rational functions

$$T_i(z) := (G' + H'B_i)(zI - AB_i)^{-1}A + H, \quad i = 1, \dots, N$$

are ESPR. The converse however is not necessarily true since P in part (ii) of Theorem 2 is unique. Finally, the following theorem generalizes Theorem 2 letting P be parameter dependent.

Theorem 3. Consider $\mathcal{B} = \text{conv}_N(B)$. If there exist matrices $P_i > 0$ for $i = 1, \dots, N$ and matrices G and H satisfying the LMI

$$\begin{bmatrix} GB_i + B_i'G' - P_i & B_i'H' - G \\ HB_i - G' & A'P_iA - H - H' \end{bmatrix} < 0, \quad i = 1, \dots, N \quad (13)$$

then the set $\mathcal{A}$ is Schur.

Proof. Considering (3) and noticing that the matrix function $P(\xi) := \sum_{i=1}^N \xi_i P_i$ is always symmetric and positive definite we multiply each inequality in (13) by ξ_i and add all terms to obtain

$$\begin{bmatrix} GB(\xi) + B(\xi)'G' - P(\xi) & B(\xi)'H' - G \\ HB(\xi) - G' & A'P(\xi)A - H - H' \end{bmatrix} < 0.$$

Multiplying the above inequality on the left by $[I \ B(\xi)']$ and on the right by its transpose we get (5) for all $\xi_i \geq 0, \sum_{i=1}^N \xi_i = 1$ which proves the proposed theorem. $\square$

This theorem provides the discrete-time counterpart of the main result in [1]. As in the continuous-time case, and in contrast to the case of simultaneous stability, the later theorem allows the LMI (13) to be satisfied by distinct matrices $P_i > 0, i = 1, \dots, N$, consequently to build the parameter-dependent Lyapunov function (4). An important remark is that, in general, the result of Theorem 3 is only sufficient, the main difficulty to establish necessity being that the basic equalities (7) (see the proof of Theorems 1 and 2) cannot be enforced unless P is assumed invariant, that is, assuming simultaneous stability. As in the continuous-time case, the existence of special uncertain models, defined by particular convex sets $\mathcal{B}$ for which necessity holds is still an open question. In this sense, notice that it is possible to show that (13) is feasible if, and only if, there exist matrices G and H such that the rational functions

$$T_i(z) := (G' + H'B_i)(zI - AB_i)^{-1}A + H, \quad i = 1, \dots, N$$

are ESPR. Notice that in this case the Lyapunov matrices P_i may be different for each $B_i, i = 1, \dots, N$ which lets us conclude that the existence of ESPR functions $T_i(z)$ as given above is sufficient for $\mathcal{A}$ to be Schur.

4. Specific problems

In this section we apply the previous results to the same two robust stability problems discussed in [1]. The first one is the D-stability problem [5]. This result is important and stands by itself once most part of the D-stability literature is concerned with continuous-time systems. The second one is the robust stability problem in the presence of diagonally structured norm bounded uncertainty, a problem in the origin of RH_∞ theory [8].

4.1. D-stability

A given matrix $A \in \mathbb{R}^{n \times n}$ is said to be discrete-time D-stable if all matrices of the form AB , with $B \in \mathcal{B}_D$ are Schur stable. No necessary and sufficient

condition for D-stability is currently available and a well-known [3] discrete-time sufficient condition for D-stability is provided by the following lemma.

Lemma 1. *Given $(A, \mathcal{B}_D)$, if matrix A is diagonally stable, i.e., the Lyapunov inequality $A'PA - P < 0$ is satisfied for some positive diagonal matrix $P = P_D > 0$ then $\mathcal{A}$ is Schur.*

As a single diagonal matrix P proves stability for the whole set $\mathcal{B}_D$, this is a simultaneous stability condition. As in the continuous-time case we will try to reduce conservativeness in this test by applying Theorem 3. In this direction and by Theorem 3, $\mathcal{A}$ is Schur whenever (13) is feasible for the extreme matrices

$$B_i = \text{diag}(\gamma_1, \dots, \gamma_j, \dots, \gamma_n) > 0, \quad i = 1, \dots, 2^{n-1}, \quad (14)$$

where $\gamma_j \in \{1, -1\}$, $j = 1, \dots, n$ and $B_k \neq -B_l \forall k, l$.

Notice that it is not necessary to test for opposite vertices, thus reducing the number of vertices from 2^n in $\mathcal{B}_D$ to 2^{n-1} . This is possible once the eigenvalues of A satisfy $\alpha\lambda_i(A) = \lambda_i(\alpha A)$, $i = 1, \dots, n$ for any scalar $|\alpha| \leq 1$, in particular $\alpha = -1$. Also remarkable is the fact that in the discrete-time case the set $\mathcal{B}_D$ given by (1) is already a closed and limited polyhedron while in the continuous-time case a special polyhedron had to be generated (see [1]). Nevertheless, the number of vertices of the discrete-time polyhedron is of order $O(2^{n-1})$ while in the continuous-time case the equivalent polyhedron has order $O(n)$. This fact illustrates one of the reasons why the D-stability problem is inherently more involved for discrete-time systems than for continuous-time systems, which, in a sense, may explain the cited preference for continuous-time results found in the literature. Another indication of this fact is that the second condition provided in Lemma 4 of [1], namely the existence of $(AP)_D$ diagonal, to the authors knowledge, does not have a discrete-time counterpart. The discrete-time counterpart of the first part of this lemma is given below and shows that the algebraic condition for D-stability in Lemma 1 is indeed covered by Theorem 3.

Lemma 2. *If Lemma 1 holds for a given matrix $A \in \mathbb{R}^{n \times n}$ then Theorem 3 with (14) also holds.*

Proof. In order to prove this lemma notice that B_i in (14) is such that $B_i'B_i = B_iB_i' = I$, $\forall i$ so that

$$A'P_D A - P_D = A'P_D A - 2P_D + B_i'P_D B_i < 0$$

for all $i = 1, \dots, 2^{n-1}$ since P_D is a diagonal matrix. Now, since $P_D > 0$ we have by Schur complement that

$$\begin{bmatrix} -P_D & B_i' P_D' \\ P_D B_i & A' P_D A - 2P_D \end{bmatrix} < 0, \quad i = 1, \dots, 2^{n-1},$$

which is exactly the matrix on the left-hand side of (13) with $G = 0$ and $H = P_i = P_D$. This proof is concluded if we notice that

$$A' P_D A - P_D < 0 \Rightarrow A' P_D A - 2P_D < 0$$

so that (13) is indeed feasible. $\square$

Although this lemma shows that our results contains a well-known result, it is very difficult to try to evaluate how larger is the subset of the discrete-time D-stable matrices parametrized by Theorem 3 and (14). However, as in the continuous-time case, we believe that this new result is simple and general enough to be useful to test a given matrix for D-stability, which will be illustrated by a numerical example in Section 5.

4.2. RH_∞ Theory

Consider now the classical robust stability problem consisting in the determination of conditions under which all real matrices of the form $F + U\Delta V \in \mathbb{R}^{n \times n}$ such that $\Delta' \Delta \leq \alpha^2 I$ are discrete-time asymptotically stable. From Schur complement we have the equivalent inequalities

$$(F + U\Delta V)' P (F + U\Delta V) - P < 0,$$

$$P > 0 \iff \begin{bmatrix} P & (F + U\Delta V)' P \\ P(F + U\Delta V) & P \end{bmatrix} > 0,$$

while completing the squares and using the bound over Δ we get

$$\begin{aligned} \begin{bmatrix} P & (F + U\Delta V)' P \\ P(F + U\Delta V) & P \end{bmatrix} &\geq \begin{bmatrix} P - V'V & F'P \\ PF & P - PU\Delta\Delta'U'P \end{bmatrix} \\ &\geq \begin{bmatrix} P - V'V & F'P \\ PF & P - \alpha^2 PUU'P \end{bmatrix}, \end{aligned}$$

which is valid for P symmetric and all $\Delta' \Delta \leq \alpha^2 I$. So the proposed problem has an affirmative answer whenever the inequality (obtained from the previous inequality by Schur complement)

$$F'(P^{-1} - \alpha^2 UU')^{-1}F - P + V'V < 0 \quad (15)$$

is feasible for some $P > 0$. In this case the rational function $G(z) := V(zI - F)^{-1}U$ has an RH_∞ norm less than α^{-1} (see [8]).

This result can be very conservative mainly if A is constrained to have some particular structure as, for instance, $A = A_D$. Another source of conservatism is the fact that there must exist a single matrix P satisfying (15), which implies that only simultaneous stability is tested. So, using Theorem 3 the stability conditions for this problem can be improved since $A_D' A_D \leq \alpha^2 I$ is a convex polyhedron defined by $N = 2^n$ matrices A_i , $i = 1, \dots, N$. Indeed, setting $A = I$ and $\mathcal{B} = \text{conv}_N(B)$ where

$$B_i = F + U A_i V, \quad i = 1, \dots, N$$

the given robust stability problem reduces to verifying whether $\mathcal{A}$ is stable. Notice that the formulation given in [1] is, in general, no longer valid for discrete-time systems since F may be singular.

5. Illustrative examples

In order to illustrate the results of this paper let us consider a simple example. Given the following matrices

$$A = \begin{bmatrix} 0.8 & -0.25 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0.2 & 0.03 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad c' = \begin{bmatrix} 0.8 \\ -0.5 \\ 0 \\ 1 \end{bmatrix}$$

we wish to determine the maximum value of $\alpha \in \mathbb{R}$ such that the matrix $A + \theta bc$ is Schur stable for all $|\theta| < \alpha$. To formulate this problem in the context of this paper we define $A = I$ and $\mathcal{B} = \text{conv}_2(B)$ with $B_1 = A + \alpha bc$, $B_2 = A - \alpha bc$ and determine the maximum value of α such that $\mathcal{A}$ remains stable. The results are shown in Table 1. One can verify that the exact maximum value of α is 0.4619. The feasibility of the LMI given by Theorem 3 has been verified using an LMI solver, also leading to the exact maximum value. Thus, the stability condition of Theorem 3 provides the best value for α . So, although only sufficient, it is not conservative in the present case. A conservative result is obtained if we impose $G = H$. It is interesting to notice that even with this hard constraint, Theorem 3

Table 1
Maximum values of α

Method	α
Exact maximum value	0.4619
Theorem 3	0.4619
Theorem 3 with $G = H$	0.4524
Theorem 2	0.4279
RH_∞	0.2956

provides better result than the simultaneous stability (Theorem 2) and RH_∞ (Section 4.2) approaches.

As another example, let us consider the following matrix

$$A = \begin{bmatrix} 0.97701 & -0.32047 & -0.089600 \\ 0.07282 & 0.66868 & 0.389559 \\ 0.15153 & -0.27886 & 0.697892 \end{bmatrix},$$

which is known to be D-stable [3]. It is also known [12] that there is no $P = P_D$ such that Lemma 1 holds. With the help of an LMI solver it is simple to verify that it is feasible with respect to the sufficient condition given by Theorem 3, i.e., the LMI (13) is feasible for (14).

6. Conclusion

In this paper we have shown how the recently obtained stability results for continuous-time systems in [1] can be fully extended to cope with discrete-time systems. These results consists of a method to establish the stability of all matrices belonging to a set $\mathcal{A} := \{AB : B \in \mathcal{B}\}$, when $A \in \mathbb{R}^{n \times n}$ and $\mathcal{B} \subset \mathbb{R}^{n \times n}$ is a known convex set. The given stability conditions are formulated as a finite set of LMI problems leading to parameter dependent Lyapunov functions which are less conservative than taking simultaneous invariant Lyapunov functions for the whole set $\mathcal{B}$. As in the continuous-time case, the key idea was to embed the original stability problem into a positive realness condition of a finite set of matrix transfer functions. Applications to particular problems, namely, D-stability and RH_∞ theory are discussed. Two examples illustrate the significant reduction of conservativeness of the proposed method with respect to other known approaches.

References

- [1] J.C. Geromel, M.C. de Oliveira, L. Hsu, LMI characterization of structural and robust stability, *Linear Algebra Appl.* 285 (1998) 69–80.
- [2] E. Kaszkurewicz, L. Hsu, A note on the absolute stability of nonlinear discrete time systems, *Internat. J. Control* 40 (1984) 867–869.
- [3] A. Bhaya, E. Kaszkurewicz, On discrete-time diagonal and D-stability, *Linear Algebra Appl.* 187 (1993) 87–104.
- [4] S.K. Persidskii, Problem on absolute stability, *Automat. Remote Control* 12 (1969) 1889–1895.
- [5] D. Hereshkowiitz, Recent directions in matrix stability, *Linear Algebra Appl.* 171 (6) (1992) 161–186.
- [6] J.C. Geromel, On the determination of a diagonal solution of the Lyapunov equation, *IEEE Trans. on Automatic Control* 30 (4) (1985) 404–406.

- [7] S.P. Boyd, L. El Ghaoui, E. Feron, V. Balakrishnan, *Linear Matrix Inequalities in System and Control Theory*, Philadelphia, PA: SIAM, 1994.
- [8] P. Colaneri, J.C. Geromel, A. Locatelli, *Control Theory and Design: An $RH_2 - RH_\infty$ viewpoint*, Academic Press, New York, 1997.
- [9] J.C. Geromel, P.L.D. Peres, J. Bernussou, On a convex parameter space method for linear control design of uncertain systems, *SIAM J. Control and Optimization* 29 (2) (1991) 381–402.
- [10] K. Premaratne, E.I. Yuri, Discrete-time positive-real lemma revisited: the discrete-time counterpart of the Kalman–Yakubovitch lemma, *IEEE Trans. Circuit and Systems – I: Fundamental Theory and Appl.* 41 (11) (1994) 747–750.
- [11] M.M. Ekanayake, K. Premaratne, E.I. Yuri, Some corrections on Discrete-time positive-real lemma revisited: the discrete-time counterpart of the Kalman–Yakubovitch lemma, *IEEE Trans. Circuits and Systems – I: Fundamental Theory and Appl.* 43 (8) (1996) 707–708.
- [12] A. Bhaya, F.C. Mota, Equivalence of stability concepts for discrete time varying systems, *Internat. J. Robust and Nonlinear Control* 4 (6) (1994) 725–740.